



Snow-line

Above a certain height, called the snow-line, the air is so cold that the snow never melts. In the tropics the snow-line is well over 5,000 m (16,000 ft), but comes down to 600 m (1,900 ft) in Greenland and is at sea-level at the North and South Poles.

Mount Kilimanjaro, Tanzania

Avalanches

The snow cover on steep slopes is often far from stable. If the layers are not well compacted, even a slight disturbance – a falling rock, a skier, or even a shout – can make an entire snowfield collapse in an avalanche.

A powder snow avalanche such as this can produce shock waves powerful enough to explode buildings.

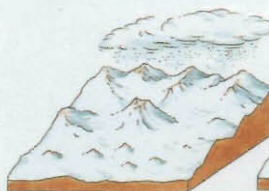


Ice sheets and caps

Ice sheets are huge layers of ice, thousands of metres thick, that may cover not just a single valley but an entire continent. Ice caps are smaller dome-shaped sheets of ice that cover a mountain. The sheets of ice over Antarctica and Greenland are also called ice caps. The ice deep within the polar ice caps first fell as snow many millions of years ago.

Formation of an ice cap

Ice caps form gradually by accumulation as snow falls, stays frozen, and is compacted by the addition of new snow. Some ice is lost by "ablation" (melting and evaporation), but if the ice is formed faster than it is lost, then the ice cap grows.



1 An ice cap forms when the snow covering a peak remains frozen all year.



2 Fresh snowfalls compact the snow beneath, turning it into dense crystals.



3 Eventually, the lower layers are compacted into solid opaque ice.

Ice sheets are thousands of metres thick, but vary in extent and depth between summer and winter, which effects the Earth's climate.

Isolated mountaintops jutting through the surface of an ice cap are called nunataks.

Glacier moves by sliding over melted ice.

Rocks under glacier are slowly eroded.

Ice fall – crevasses form where glacier flows over steep rock.

Around 10,000 icebergs a year break away from the glaciers in Greenland.

When Arctic glaciers reach the sea, the tides and waves heave the ice up and down, cracking bits off to float away as icebergs, a process known as calving.

Only about 12 per cent of an iceberg is visible above the surface of the ocean.

Icebergs

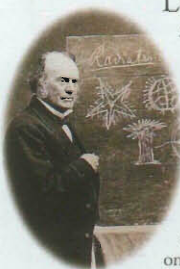
Icebergs are huge chunks of ice that have broken off from the edge of an ice sheet or glacier to float in the sea. They are generally rounded or block-like in shape. Icebergs float because ice is less dense than water, but it is only a little less dense, so about one-eighth of the iceberg is visible above the surface.

Icebergs may be broad and tabular (flat). They are often hundreds of kilometres long and may last for years before melting.

It is estimated that the average age of the ice in an iceberg is 5,000 years.

Louis Agassiz

Swiss-American geologist Louis Agassiz (1807–73) realized that past ice ages had shaped the landscape. In 1836, he noted that glaciers are not static, but move, and found rocks that had been scoured by glaciers. He concluded much of northern Europe had at one time been covered by ice.



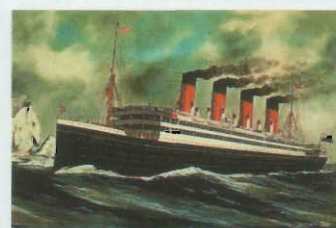
Shaded areas show the extent of the ice cover during the last ice age.

Ice Age

There is no doubt that ice ages have occurred several times in the Earth's past. Some geologists believe they are linked to the variations in the energy reaching Earth from the Sun as the Earth wobbles and tilts in its orbit. Others think there may be some other trigger for an ice age.

Titanic disaster

Because most of an iceberg is hidden below the surface, it can pose a real hazard to shipping if one drifts across sea lanes. In 1912, the luxury liner *Titanic* sank after a collision with an iceberg, resulting in the loss of about 1,500 lives.



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